Voronoi and Graph Entropy in Multicomponent Charged Liquids

Work plan for an independent structural assessment

EntropicFluids collaboration

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Abstract

This plan proposes a staged analysis of existing molecular-dynamics trajectories: first local Voronoi topology and chemical neighborhoods, then, if sampling permits, cluster-entropy extrapolation. Thermodynamic integration (TI) remains the reference for excess entropy. The objectives are to determine whether structural diversity tracks TI entropy, whether environments around unchanged divalent ions explain their mobility, and whether adding physically distinct components has effects beyond adding species labels. No additional MD is required for the initial pilot. The protocol below is proposed work, not a report of completed Voronoi calculations.

1 Scientific questions and prior work

Rosenfeld scaling relates reduced transport to excess entropy [1]. Charged-mixture applications precede the present study [2]. Generalized tracer scaling has been tested in model mixtures, with limitations when component mobilities decouple [3]. More recent mixture modeling uses entropy together with limiting transport properties and mixing rules; one bulk entropy is not generally a unique predictor of each diffusion coefficient [4].

For electrolytes, Wang et al. connect multiple salts, altered solvation, and enhanced lithium mobility [5]. He et al. explicitly report a Rosenfeld relation for lithium self-diffusion in multi-anion electrolytes [6]. These motivate comparison, but do not establish a universal causal effect of component count. Only the abstract and bibliographic record of Ref. [6] were accessible during preparation of this plan; its entropy estimator and controls must be checked in the full paper before making a detailed methodological comparison.

Two graph approaches should be distinguished:

1. **Local cell topology.** Yoon et al. [7] classify Voronoi cells by their complete graph topology, using Weinberg vectors, and compare topological Shannon entropy with TI entropy and transport in repulsive soft-sphere fluids. Their entropy conversion is interaction dependent; rare types and correlations between adjacent cells matter. This is the closest precedent for the initial pilot. The underlying topology framework and an implementation are described in Refs. [8, 9].
2. **Cluster entropy.** Zhou and Milner [10] classify Voronoi-derived clusters by graph isomorphism and estimate structural entropy per particle from the growth of subsystem entropy with particle number. They benchmark hard-particle systems against previous configurational-entropy estimates from TI. This motivates the more expensive second stage, but does not prove equivalence to excess entropy in our attractive, charged mixtures.

Voronoi entropy has also been used for hydrogen-bonded liquids and binary mixtures [11]; that application illustrates structural characterization rather than a general absolute-entropy prescription.

The proposed contribution is a benchmark across controlled charged mixtures, with an exact label control and independent TI and diffusion data. The literature search establishes precedent, not priority for every element of this design.

2 Entropy definitions and comparison rules

The reference is classical excess entropy per particle relative to the ideal mixture at the same temperature, density, and composition:

$$s_{\mathrm{ex}} = \frac{S - S_{\mathrm{id,mix}}}{N}, \qquad \frac{s_{\mathrm{ex}}}{k_B} = \frac{U_{\mathrm{ex}}}{N k_B T} - \frac{A_{\mathrm{ex}}}{N k_B T}. \tag{1}$$

Ideal mixing, $-k_B \sum_i x_i \ln x_i$, is excluded. Do not substitute the excess entropy of mixing relative to pure liquids for this quantity.

For a chosen local topological class α, define the dimensionless information entropy

$$H_{\mathrm{V}} = -\sum_{\alpha} p_\alpha \ln p_\alpha. \tag{2}$$

This describes the diversity of a coarse structural representation. It is not automatically s_{ex}/k_B, and summing single-cell entropies omits correlations. Likewise, glass configurational entropy, which counts basins after separating vibrational motion, is a different target from our full liquid excess entropy.

Use a simulated ideal-gas point ensemble with the same periodic geometry and the identical estimator to define an optional topological reference $\Delta H_{\mathrm{V}} = H_{\mathrm{V}} - H_{\mathrm{V,id}}$. Do not assume that subtracting this reference makes it a thermodynamic excess entropy. If chemical colors are used, match their composition in the reference. Any fitted mapping to TI must be explicitly called a calibration and tested on excluded compositions.

The completed classical 2PT assessment follows Refs. [12, 13]. Its internal checks passed, but its absolute bias varies with composition and it overestimates one controlled TI increment. Keep it as a secondary comparator. In particular, diffusion enters its spectral partition, whereas the proposed static graph descriptors do not use velocities.

3 Systems, existing evidence, and priorities

All states have $N = 4000$, $T^* = 2$, $\rho^* = 0.7$, $x_{+2} = x_{-2} = 0.03$, unit masses and nominal diameters, and three replicas. Ions and primary neutral solvent have $\varepsilon = 1$. Unlike LJ amplitudes follow geometric mixing. The shifted LJ cutoff is fixed at 2.5σ throughout TI; electrostatics use PPPM at tolerance 10^{-5}. The parameter $I = \frac{1}{2}\sum_i x_i q_i^2$ below is dimensionless; it is not a molar ionic strength.

Table 1: Composition map. Secondary fractions and amplitudes correspond in order. The monovalent fraction is for each sign; all rows also contain 3% of each divalent sign.

Code	Primary x	Secondary x	Secondary ε	$x_{\pm1}$	I
B3	.94	—	—	0	.12
B5	.84	—	—	.05	.17
U4	.64	.30	.25	0	.12
L5	.64	.15+.15	.25, .25	0	.12
M5	.64	.15+.15	.0625, .5625	0	.12
H5	.64	.15+.15	.01, .81	0	.12
U6	.54	.30	.25	.05	.17
H7	.54	.15+.15	.01, .81	.05	.17

1. **Mandatory null: U4/L5.** Identical interactions, with one secondary solvent split into two labels. Formal ideal mixing changes by $0.30 \ln 2 = 0.20794$ per particle in units of k_B; diffusion is identical in the completed trajectories. TI differences are consistent with sampling noise. Verify positional identity by ID rather than assuming it from equal diffusion. Also create a synthetic label split of the exact same snapshots to guarantee an exact implementation test.

2. **Primary contrasts: H5–U4 and H7–U6.** Both TI increments are approximately $+0.1033k_B$. Primary-window probe diffusion increases by 4.81% and 5.20%, respectively. The former loses nominal significance in the late MSD window; the latter is unresolved across replicas. These small effects must not be described as established merely because a structural descriptor changes significantly.
3. **Moderate perturbation: M5–U4.** TI increases by $0.0304k_B$; the approximately 1.5% probe response is unresolved. Include this case to avoid selecting only large structural changes.
4. **Broad references: B3/U4/H5 and B5/U6/H7.** Baseline-to-design probe increases are about 20% and 18%. These comparisons change mean interaction strength and supply range, rather than an isolated component-count test. B3/B5 also changes ionic strength and is a secondary contrast.

U4/L5/M5/H5 have identical $\sum_i x_i\sqrt{\varepsilon_i} = 0.85$ and mean bare pair amplitude $\sum_{ij} x_i x_j \varepsilon_{ij} = 0.7225$; U6/H7 also share these values. Actual energy, effective excluded volume, solvation, and friction need not match. L5 is a duplicate control, not another independent state in regressions. The pilot should cover all eight states if processing is inexpensive; otherwise begin with U4/L5/H5/U6/H7 and B3, then add M5/B5 before drawing study-wide conclusions.

4 Data handover and reproducibility

Paths are relative to the repository root; raw trajectories remain local and are not included in the Git repository. Supply them separately to the analyst.

- `ComponentEntropyStudy_20261030/selected.json`: exact species definitions and case names.
- `ComponentEntropyStudy_20261030/md/<case>/replica_<r>/transport/positions.nc`: all-particle, unwrapped positions every 0.5τ over 1000τ production. Read stored IDs, types, frame times, and box information explicitly; do not infer atom ordering from row number.
- Corresponding `transport/data.mixture`, `in.run`, and replica `manifest.json`: charges/type mapping, box, inputs, and provenance.
- `results/summary.json`, `results/diagnostics.json`, and `results/conclusions.md` under `ComponentEntropyStudy_20261030/`: TI and transport comparisons, intervals, estimator sensitivities, and interpretation.
- `md/<case>/replica_<r>/replica_result.json` and `transport/msd_multi_origin.csv`: replica-level reference values.
- `ComponentEntropyStudy_20261030/two_pt_rehabilitation/`: corrected 2PT methods, replica results, `summary.json`, and `report.md`.

Case directories are B3: `baseline_3C`; B5: `mono_f010`; U4: `weak_f030_e025`; L5: `dispersion_split_d000`; M5: `dispersion_split_d025`; H5: `dispersion_split_d040`; U6: `mono_weak_uniform6C`; H7: `mono_weak_split7C`.

Record source commit, checksums, tool versions, all tessellation/graph options, selected frame IDs, random seeds, and estimator settings. Retain canonical graph identifiers and type counts, not just entropy estimates. Write new outputs to a separate directory such as `VoronoiEntropyAssessment/`. Use the colleague's validated tools; the scientific definitions below should be agreed before comparing descriptor values with diffusion.

5 Stage A: local structural pilot

5.1 Geometry and graph validation

Wrap coordinates into each frame's periodic box before tessellation; preserve the unwrapped record for displacement analysis. Tessellate *all* particles, then condition statistics on the central species. Tessellating only ions would measure a different neighborhood and is not the primary protocol. Equal nominal diameters motivate ordinary Voronoi cells initially. Weighted cells, if used later, require an explicit radius convention and sensitivity analysis.

Validate periodic cells by checking that volumes sum to the box volume, adjacency is reciprocal, and results are invariant under translation, atom ordering, and type-code renaming. Distinguish the edge graph of a single polyhedron from a particle-adjacency graph. Canonicalization must resolve graph isomorphism exactly; a hash or face-count vector alone is not a complete topological identifier. Check any treatment of tiny faces or near-degenerate vertices on a fixed subset before selecting a primary convention.

Use instantaneous configurations as primary. Quenching is an optional, separately identified inherent-structure analysis; it changes the target and must not silently replace liquid configurations.

5.2 Descriptors and controls

Produce the following for the full liquid and separately for $+2$, -2, and each mechanically distinct neutral species:

1. Full uncolored cell-topology frequencies and H_{V}; retain face-count and coordination distributions as simpler diagnostic summaries.
2. Cell volume, surface area, and shape distributions. These capture metric information discarded by topology; report their moments without treating a bin-dependent differential entropy as an absolute thermodynamic entropy.
3. Neighbor composition and charge distributions around the central ion. If supported, calculate colored neighborhood graph classes. Specify precisely whether colors belong to faces (neighbor particles) or nodes of a particle graph, and canonicalize colors consistently with graph automorphisms.
4. Geometry-only, charge-only, and full mechanical-species versions, with mechanically identical L5 labels merged in the physical-species version. Also report label-resolved L5 as a deliberate bookkeeping diagnostic.

At fixed coordinates, permute species labels while preserving counts and holding the selected central ion identity fixed for ion-centered statistics. Use multiple seeded permutations, preferably at least 100 for the pilot. This is a chemical-association null, not an equilibrium realization of the charged Hamiltonian. Compare actual and randomized conditional neighborhood distributions to expose ordering beyond composition. Keep purely combinatorial color diversity separate from geometry. Entropy differences against permutation controls are structural statistics, not automatically thermodynamic contributions.

5.3 Sampling and uncertainty

Start with 100 frames evenly spaced over each dense production trajectory; repeat with 200 and 400 frames if available and needed. Increase separation or use blocks after measuring correlation of representative topology and coordination observables. All 4000 particles can contribute to bulk statistics, but only 120 ions per sign occur per frame and neighboring cells are correlated.

Report raw entropy, bias-corrected estimates supported by the tools, singleton fractions, discovery curves, and rare-type coverage. An unseen-type estimator does not create independent samples. Repeat with matched sample sizes and thinning. If entropy continues to grow with sample count, report an unresolved estimate rather than selecting the apparent plateau. Include independent ideal-gas point ensembles with the same pipeline; their topology can require much more sampling than the dense liquid.

Keep replicas separate throughout estimation. Use paired differences across the three replica indices for composition contrasts; report nominal 95% Student-t intervals with two degrees of freedom. Block resampling within replicas diagnoses temporal uncertainty but does not turn cells, overlapping clusters, frames, or permutations into independent MD replicas. Check first and second trajectory halves for stationarity. Monitor cation/anion symmetry; the existing baseline transport asymmetry is a sampling flag.

6 Stage B: thermodynamic and transport assessment

Compare descriptor *increments* first, particularly H5–U4 and H7–U6. Then compare absolute-state ordering against TI, HNC, and corrected 2PT. Calculate the independent pair contribution from species RDFs where feasible:

$$\frac{s_2}{k_B} = -2\pi\rho \sum_{ij} x_i x_j \int_0^\infty [g_{ij}(r) \ln g_{ij}(r) - g_{ij}(r) + 1] r^2 \, dr. \tag{3}$$

The sum is over ordered species pairs. Document finite-box cutoff, binning, normalization, and tail sensitivity; merging identical labels must preserve the result. This gives a simple structural baseline before crediting graph analysis with additional predictive information.

For transport use the existing common-system-COM MSD estimates:

$$D_{\text{net}} = \sum_i x_i D_i, \quad D_{\text{probe}} = \tfrac{1}{2}(D_{+2} + D_{-2}), \quad D_i^{\text{R}} = D_i \rho^{1/3} \sqrt{m_i/(k_B T)}. \tag{4}$$

Use total number density in the reduction. At common T, ρ, and mass, diffusion ratios are unchanged by it. Neither D_{net} nor D_{probe} is conductivity, mutual diffusion, or diffusion of a bound neutral ion pair. Retain the primary 20–100τ MSD fit and the 10–50 and 50–200τ sensitivity windows.

Compare $\ln D_i^{\text{R}}$ with bulk topology and ion-centered descriptors. Keep fixed-I families separate; exclude L5 from fit counts. With only seven physical states, a high training R^2 is weak evidence. Limit models to one descriptor plus an intercept initially; show residuals, paired contrasts, leave-one-state-out errors, and transfer between the two salt backgrounds. Do not select descriptors or graph parameters by maximizing agreement with diffusion and then present that agreement as an independent test.

As an optional mechanistic extension, relate an ion's initial environment to its subsequent squared displacement at fixed lags (e.g., 5, 20, and 50τ), with the same whole-system COM correction as the reference analysis. Call this conditional mobility, not a local self-diffusion coefficient. Analyze within each state so that between-state differences cannot create a spurious local association. Neighbor persistence may help distinguish geometric packing from chemical residence effects; the 0.5τ output spacing limits short-time transition and residence-time resolution.

7 Stage C: conditional cluster-entropy extrapolation

Proceed only after Stage A passes the label and convergence checks and the analyst demonstrates adequate graph-count coverage. First reproduce a small hard-particle benchmark using the cluster construction and entropy convention of Zhou and Milner [10]. Their full method must be consulted for cluster selection, boundary treatment, and corrections; the schematic formula below is not a replacement implementation.

For a fixed, documented rule for selecting connected n-particle neighborhoods, estimate canonical graph-class probabilities and

$$H_G(n) = -\sum_g p_n(g) \ln p_n(g), \qquad h_G(n) = H_G(n+1) - H_G(n). \tag{5}$$

Begin with an adaptive sequence such as $n = 2, 3, 4, 5, 6, 8, 10$, extending only while class coverage remains credible. Rooted and unrooted graphs, graph-radius and nearest-n neighborhoods, and induced versus boundary-completed graphs define different distributions: select one primary convention and document all alternatives. Preserve periodic adjacency before extracting local clusters. Distinguish clusters sampled around probes from bulk clusters; their entropy slopes have different conditioning.

Evaluate uncolored graphs first, followed by charge/species colors with merged identical labels. Apply the same ideal-gas and fixed-composition permutation references. Check sample-size growth for every n: saturation of observed entropy at roughly the logarithm of the number of sampled clusters is a coverage failure, not physical convergence.

A defensible extensive regime would have stable $h_G(n)$ or stable fitted slopes under changes of fit range, trajectory half, sample size, and replica. Test boundary corrections rather than presuming exact linearity for small clusters. Require at least four consecutive, adequately sampled integer sizes for a provisional slope fit. If a converged regime cannot be reached before rare graphs dominate, stop and report finite-cluster information statistics. Do not force an infinite-size extrapolation.

Even a stable slope is entropy of the chosen graph representation. In an energetic liquid, configurations with the same graph can have different distances, energies, and vibrational phase-space volumes. The missing within-class information must be accounted for, or the result retained as structural entropy. A TI calibration is an empirical mapping and requires held-out tests. It does not establish an exact statistical-mechanical identity.

8 Decisions, deliverables, and follow-up

Checkpoint	Required evidence and action
Implementation	Periodicity and invariance checks pass; a synthetic label split leaves geometry and merged-color results identical to numerical precision. Audit actual U4/L5 snapshots before interpreting any discrepancy.
Sampling	Discovery curves, thinning, halves, and replica analyses are stable at the scale of the target contrasts. Otherwise extend postprocessing samples or report unresolved descriptors; add MD only if available sampling is inadequate.
Thermodynamic proxy	Report raw structural results independently. If a calibrated s_{ex} predictor is proposed, use provisional practical targets of $0.02k_B$ error in controlled increments and $0.05k_B$ in absolute values, with held-out validation. These are project targets, not literature guarantees; wide intervals do not establish equivalence.
Mechanistic value	Establish whether charge/species-aware probe descriptors explain behavior beyond bulk topology, TI, or s_2. An unresolved 5% transport contrast remains unresolved even if its structural contrast is clear.
Cluster extension	Attempt extrapolation only with validated construction, adequate class coverage, and a stable entropy-growth regime. No stable regime means no reported thermodynamic entropy from extrapolation.

The handover should contain a machine-readable per-state/per-replica table; paired contrast tables with intervals; graph and chemical-null definitions; sample-coverage and estimator-sensitivity plots; TI–topology and diffusion–descriptor comparisons; and, if attempted, $H_G(n)$, increments, fit-range sensitivity, and explicit extrapolation failures. Include scripts, tool versions, frame manifests, and enough graph frequencies to reproduce the entropy estimates. A short written assessment should distinguish thermodynamic prediction, structural association, and mechanistic interpretation.

The initial effort is postprocessing of existing trajectories. Time the first few frames and graph sizes before estimating total CPU/memory cost. Local cell classification and chemical-null analysis come first; cluster enumeration is the potentially expensive part. No new GPU submission is part of this plan. If structure is well resolved but transport is not, prioritize additional replicas,

longer dense trajectories, and a second box size at fixed physical state. Higher concentrations, charge asymmetry, and different nominal sizes are later transfer tests; size changes also require a documented ordinary versus weighted Voronoi comparison.

References

[1] Y. Rosenfeld, "Relation between the transport coefficients and the internal entropy of simple systems," *Phys. Rev. A* **15**, 2545–2549 (1977). doi:10.1103/PhysRevA.15.2545.

[2] Y. Rosenfeld, E. Nardi, and Z. Zinamon, "Corresponding States Hard-Sphere Model for the Diffusion Coefficients of Binary Dense-Plasma Mixtures," *Phys. Rev. Lett.* **75**, 2490 (1995). doi:10.1103/PhysRevLett.75.2490.

[3] W. P. Krekelberg, M. J. Pond, G. Goel, V. K. Shen, J. R. Errington, and T. M. Truskett, "Generalized Rosenfeld scalings for tracer diffusivities in not-so-simple fluids: Mixtures and soft particles," *Phys. Rev. E* **80**, 061205 (2009). doi:10.1103/PhysRevE.80.061205; accessible preprint.

[4] S. Schmitt, H. Hasse, and S. Stephan, "Entropy scaling for diffusion coefficients in fluid mixtures," *Nature Communications* **16**, 2611 (2025). doi:10.1038/s41467-025-57780-z.

[5] Q. Wang, C. Zhao, J. Wang, Z. Yao, S. Wang, S. G. H. Kumar, S. Ganapathy, S. Eustace, X. Bai, B. Li, and M. Wagemaker, "High entropy liquid electrolytes for lithium batteries," *Nature Communications* **14**, 440 (2023). doi:10.1038/s41467-023-36075-1.

[6] L. He, Y. Lin, J. Li, H. Su, H. Liu, C. Yang, and C. Lian, "Bridging thermodynamics and ion transport in lithium batteries via molecular design of high-entropy electrolytes," *Molecular Systems Design & Engineering* **11**, 681–690 (2026). doi:10.1039/D6ME00060F. First published 19 May 2026; abstract-level comparison in this plan.

[7] T. J. Yoon, M. Y. Ha, E. A. Lazar, W. B. Lee, and Y.-W. Lee, "Topological extension of the isomorph theory based on the Shannon entropy," *Phys. Rev. E* **100**, 012118 (2019). doi:10.1103/PhysRevE.100.012118; accessible preprint.

[8] E. A. Lazar, J. Han, and D. J. Srolovitz, "A topological framework for local structure analysis in condensed matter," *Proceedings of the National Academy of Sciences USA* **112**, E5769–E5776 (2015). doi:10.1073/pnas.1505788112; accessible preprint.

[9] E. A. Lazar, "VoroTop: Voronoi Cell Topology Visualization and Analysis Toolkit," *Modelling and Simulation in Materials Science and Engineering* **26**, 015011 (2018; published online December 2017). doi:10.1088/1361-651X/aa9a01; accessible preprint; software and documentation.

[10] Y. Zhou and S. T. Milner, "Structural entropy of glassy systems from graph isomorphism," *Soft Matter* **12**, 7281–7288 (2016). doi:10.1039/C6SM01355D. Primary source for the cluster-entropy extension.

[11] Abdulkareem U and V. Madhurima, "Understanding complexity of hydrogen-bonded liquids through Voronoi entropy," *Chemical Physics Impact* **8**, 100403 (2024; available online 30 November 2023). doi:10.1016/j.chphi.2023.100403.

[12] T. Sun, J. Xian, H. Zhang, Z. Zhang, and Y. Zhang, "Two-phase thermodynamic model for computing entropies of liquids reanalyzed," *J. Chem. Phys.* **147**, 194505 (2017). doi:10.1063/1.5001798; accessible preprint.

[13] P.-K. Lai, C.-M. Hsieh, and S.-T. Lin, "Rapid determination of entropy and free energy of mixtures from molecular dynamics simulations with the two-phase thermodynamic model," *Phys. Chem. Chem. Phys.* **14**, 15206–15213 (2012). doi:10.1039/C2CP42011B.